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BATCH 39

## AI ASSISTED CODING -14.1

### Task 1 – Portfolio Website Design

You are building a personal portfolio website to showcase your work.

Requirements:

- Create sections for About Me, Projects, and Contact.
- Use AI to:
  - o Suggest color palettes and typography.
  - o Create a responsive layout with Grid/Flexbox.
  - o Add smooth scrolling navigation.

**Prompt:** "Act as a Senior Frontend Developer. Create a single-page professional portfolio for Humera Nuzhat, a CSE student at SR University specializing in AI/ML."

- **Sections:** About Me, Projects (Web Dev, Python, DBMS), and Contact.
- **Design:** Use a 'Midnight Tech' theme with Slate (#0f172a) and Indigo (#6366f1).
- **Layout:** Use CSS Flexbox for the nav and CSS Grid for the project cards.
- **Interactions:** Add smooth scrolling and a scroll-reveal effect using JavaScript."

```
• <!DOCTYPE html>
• <html lang="en">
• <head>
•   <meta charset="UTF-8">
•   <meta name="viewport" content="width=device-width, initial-scale=1.0">
•   <title>Humera Nuzhat | AI & ML Portfolio</title>
•   <link href="https://cdn.jsdelivr.net/npm/remixicon@3.5.0/fonts/remixicon.css"
rel="stylesheet">
•
•   <style>
•     /* --- AI-SUGGESTED TYPOGRAPHY & COLOR PALETTE --- */
•     :root {
•       --bg-dark: #0f172a;      /* Deep Slate */
•       --bg-card: #1e293b;      /* Lighter Slate for cards */
•       --accent-glow: #6366f1;   /* Electric Indigo */
•       --text-main: #f8fafc;     /* Off-white */
```

```

•      --text-dim: #94a3b8;      /* Muted blue-grey */
•      --white: #ffffff;
•
•    }
•
•    * {
•      margin: 0;
•      padding: 0;
•      box-sizing: border-box;
•      font-family: 'Segoe UI', Roboto, Helvetica, Arial, sans-serif;
•    }
•
•    html {
•      scroll-behavior: smooth;
•    }
•
•    body {
•      background-color: var(--bg-dark);
•      color: var(--text-main);
•      line-height: 1.6;
•    }
•
•    /* --- RESPONSIVE LAYOUT (FLEXBOX/GRID) --- */
•    .container {
•      max-width: 1100px;
•      margin: 0 auto;
•      padding: 0 2rem;
•    }
•
•    section {
•      padding: 80px 0;
•    }
•
•    .section-title {
•      font-size: 2.2rem;
•      text-align: center;
•      margin-bottom: 3rem;
•      color: var(--white);
•    }
•
•    .section-title span {
•      color: var(--accent-glow);
•    }
•
•    /* --- NAVIGATION --- */
•    header {
•      position: sticky;
•      top: 0;
•      background: rgba(15, 23, 42, 0.9);
•      backdrop-filter: blur(12px);
•      z-index: 1000;

```

```

•         border-bottom: 1px solid rgba(255,255,255,0.05);
•     }
•
•     nav {
•         display: flex;
•         justify-content: space-between;
•         align-items: center;
•         height: 70px;
•     }
•
•     .logo {
•         font-size: 1.5rem;
•         font-weight: 800;
•         letter-spacing: -1px;
•         text-decoration: none;
•         color: var(--white);
•     }
•
•     .logo span { color: var(--accent-glow); }
•
•     .nav-links {
•         display: flex;
•         list-style: none;
•         gap: 2.5rem;
•     }
•
•     .nav-links a {
•         text-decoration: none;
•         color: var(--text-dim);
•         font-weight: 500;
•         transition: 0.3s;
•     }
•
•     .nav-links a:hover {
•         color: var(--accent-glow);
•     }
•
•     /* --- HERO SECTION --- */
•     .hero {
•         height: 70vh;
•         display: flex;
•         flex-direction: column;
•         justify-content: center;
•         align-items: center;
•         text-align: center;
•         background: radial-gradient(circle at center, #1e1b4b 0%, #0f172a
100%);
•     }
•
•     .hero h1 {

```

```

    font-size: 4rem;
    margin-bottom: 10px;
    background: linear-gradient(to right, #fff, #6366f1);
    -webkit-background-clip: text;
    -webkit-text-fill-color: transparent;
}

.hero p {
    font-size: 1.2rem;
    color: var(--text-dim);
    max-width: 600px;
    margin-bottom: 30px;
}

.btn {
    background: var(--accent-glow);
    color: var(--white);
    padding: 12px 32px;
    border-radius: 8px;
    text-decoration: none;
    font-weight: 600;
    transition: 0.3s;
    display: inline-block;
}

.btn:hover {
    transform: translateY(-3px);
    box-shadow: 0 10px 20px rgba(99, 102, 241, 0.3);
}

/* --- ABOUT SECTION --- */
.about-box {
    background: var(--bg-card);
    padding: 40px;
    border-radius: 20px;
    border: 1px solid rgba(255,255,255,0.05);
}

.about-box p {
    margin-bottom: 20px;
    font-size: 1.1rem;
    color: var(--text-dim);
}

.about-box strong {
    color: var(--white);
}

/* --- PROJECTS GRID --- */
.project-grid {

```

```

•         display: grid;
•         grid-template-columns: repeat(auto-fit, minmax(300px, 1fr));
•         gap: 2rem;
•     }
•
•     .project-card {
•         background: var(--bg-card);
•         padding: 30px;
•         border-radius: 16px;
•         transition: 0.4s;
•         border: 1px solid transparent;
•     }
•
•     .project-card:hover {
•         border-color: var(--accent-glow);
•         transform: translateY(-10px);
•     }
•
•     .project-card i {
•         font-size: 2.5rem;
•         color: var(--accent-glow);
•         margin-bottom: 15px;
•         display: block;
•     }
•
•     .project-card h3 {
•         margin-bottom: 12px;
•         color: var(--white);
•     }
•
•     .project-card p {
•         font-size: 0.95rem;
•         color: var(--text-dim);
•     }
•
•     /* --- CONTACT SECTION --- */
•     .contact-content {
•         display: flex;
•         flex-direction: column;
•         align-items: center;
•         gap: 20px;
•     }
•
•     .contact-info {
•         display: flex;
•         gap: 30px;
•         margin-bottom: 20px;
•         flex-wrap: wrap;
•         justify-content: center;
•     }

```

```

    .info-item {
        display: flex;
        align-items: center;
        gap: 10px;
        color: var(--text-main);
        font-size: 1.1rem;
    }

    .info-item i { color: var(--accent-glow); font-size: 1.3rem; }

    .social-links {
        display: flex;
        gap: 20px;
    }

    .social-links a {
        color: var(--white);
        font-size: 1.5rem;
        transition: 0.3s;
    }

    .social-links a:hover {
        color: var(--accent-glow);
        transform: scale(1.2);
    }

    /* --- RESPONSIVE DESIGN --- */
    @media (max-width: 768px) {
        .hero h1 { font-size: 2.8rem; }
        .nav-links { display: none; } /* Simplified for lab */
        .project-grid { grid-template-columns: 1fr; }
    }
</style>
</head>
<body>

    <header>
        <div class="container">
            <nav>
                <a href="#" class="logo">Humera<span>.</span></a>
                <ul class="nav-links">
                    <li><a href="#home">Home</a></li>
                    <li><a href="#about">About</a></li>
                    <li><a href="#projects">Projects</a></li>
                    <li><a href="#contact">Contact</a></li>
                </ul>
            </nav>
        </div>
    </header>

```

```
<section id="home" class="hero">
  <div class="container">
    <h1>Humera Nuzhat</h1>
    <p>CSE Student | Specializing in AI & Machine Learning | SR
University</p>
    <a href="#projects" class="btn">View My Portfolio</a>
  </div>
</section>

<section id="about" class="container">
  <h2 class="section-title">About <span>Me</span></h2>
  <div class="about-box">
    <p>
      I am a <strong>Computer Science Engineering student</strong> at
<strong>SR University</strong>, specializing in the dynamic fields of
<strong>Artificial Intelligence and Machine Learning</strong>.
    </p>
    <p>
      Beyond standard academics, I focus on the intersection of data
science and software engineering. I have a keen interest in developing predictive
models and high-performance applications that solve real-world problems. My goal
is to leverage my technical foundation in DBMS, OS, and C programming to build
next-generation intelligent systems.
    </p>
  </div>
</section>

<section id="projects" class="container">
  <h2 class="section-title">My <span>Projects</span></h2>
  <div class="project-grid">
    <div class="project-card">
      <i class="ri-html5-line"></i>
      <h3>Smart Task Manager</h3>
      <p>A sleek productivity tool developed using <strong>HTML, CSS,
and JS</strong>. Implements local storage and event listeners for a seamless user
experience.</p>
    </div>
    <div class="project-card">
      <i class="ri-terminal-box-line"></i>
      <h3>Trend Predictor</h3>
      <p>Utilized <strong>Python</strong> and Scikit-learn to analyze
datasets and forecast market trends, focusing on data cleaning and feature
engineering.</p>
    </div>
    <div class="project-card">
      <i class="ri-database-2-line"></i>
      <h3>Inventory DBMS</h3>
    </div>
  </div>
</section>
```

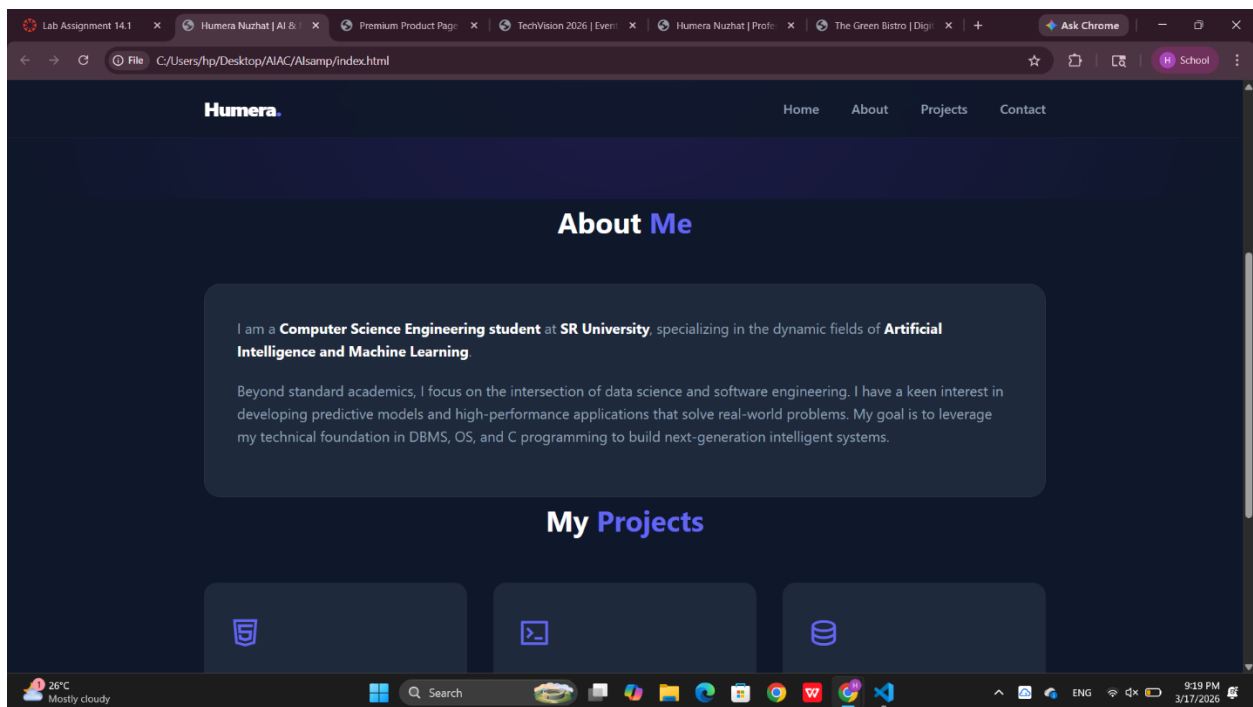
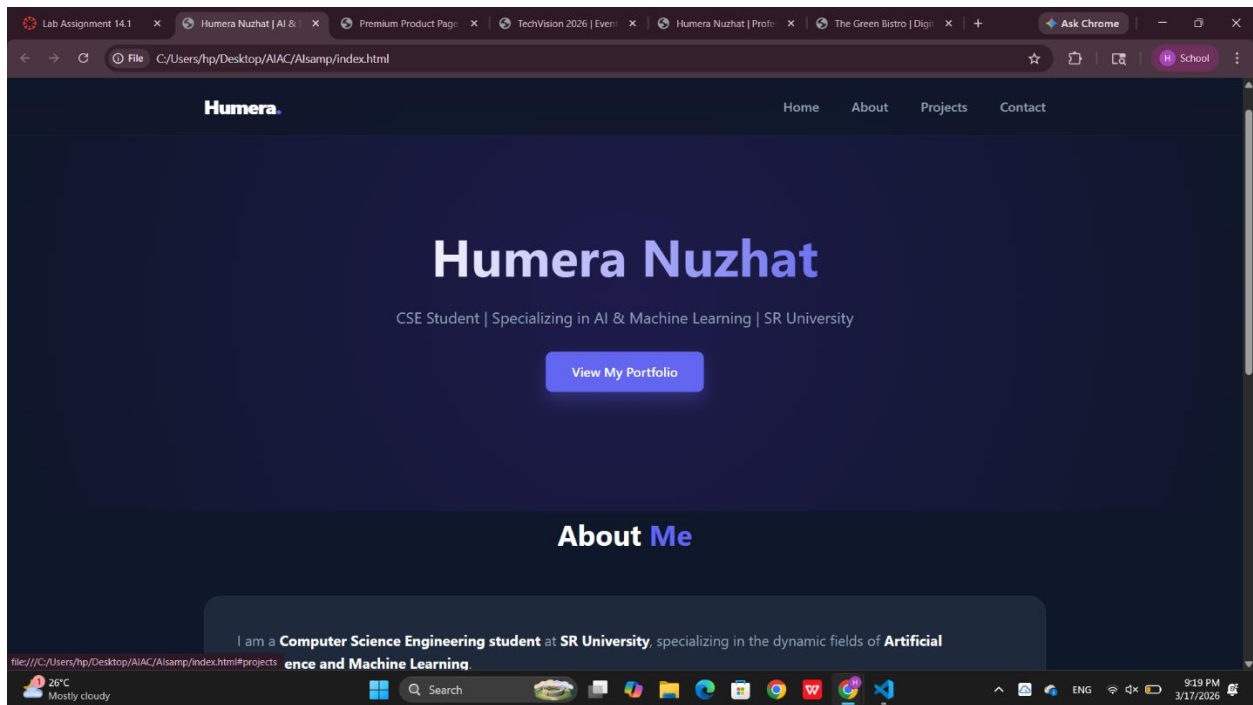
```

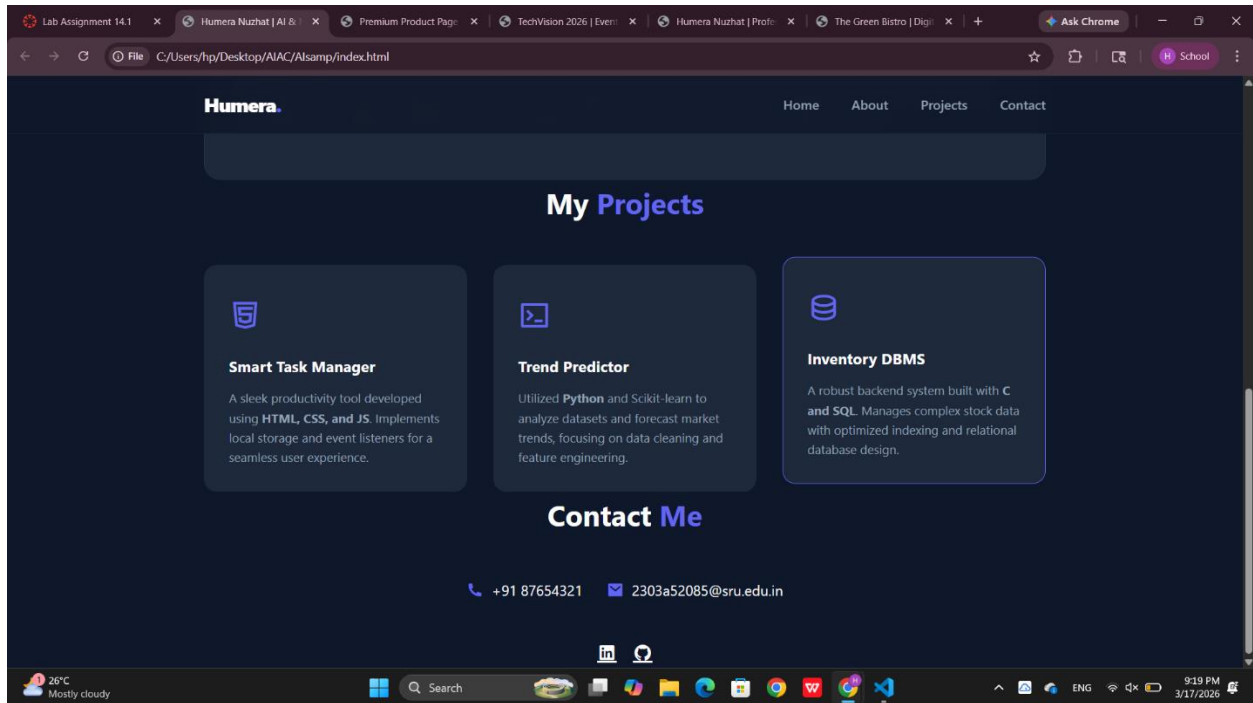
•         <p>A robust backend system built with <strong>C and SQL</strong>.
Manages complex stock data with optimized indexing and relational database
design.</p>
•         </div>
•     </div>
• </section>
•
•     <section id="contact" class="container">
•         <h2 class="section-title">Contact <span>Me</span></h2>
•         <div class="contact-content">
•             <div class="contact-info">
•                 <div class="info-item">
•                     <i class="ri-phone-fill"></i> +91 87654321
•                 </div>
•                 <div class="info-item">
•                     <i class="ri-mail-fill"></i> 2303a52085@sru.edu.in
•                 </div>
•             </div>
•             <div class="social-links">
•                 <a href="https://linkedin.com/in/humera-nuzhat-fake"><i class="ri-
linkedin-box-fill"></i></a>
•                 <a href="https://github.com/humera-nuzhat-fake"><i class="ri-
github-fill"></i></a>
•             </div>
•         </div>
•     </section>
•
•     <script>
•         // Smooth scrolling for navigation links
•         document.querySelectorAll('.nav-links a, .btn').forEach(anchor => {
•             anchor.addEventListener('click', function(e) {
•                 e.preventDefault();
•                 const targetId = this.getAttribute('href');
•                 document.querySelector(targetId).scrollIntoView({
•                     behavior: 'smooth'
•                 });
•             });
•         });
•     </script>
•
• </body>
• </html>

```

**OUTPUT:**







## Task 2 – Online Store Product Page

Design a product display page for an online store.

Requirements:

- Display product image, title, price, and "Add to Cart" button.
- Use AI to:
  - o Style with BEM methodology.
  - o Make layout responsive.
  - o Add hover effects and "Add to Cart" alert

**Prompt:** "Design a premium e-commerce product display page.

- **Elements:** Include a high-quality product image, title, price, and 'Add to Cart' button.
- **Methodology:** Use the **BEM (Block Element Modifier)** naming convention for all CSS classes.
- **Responsiveness:** Use a media query to stack the image and text vertically on mobile.
- **JS:** Add an event listener to the button that shows a success alert when clicked.
- **Effects:** Add a subtle scale-up hover effect on the product image."
- `<!DOCTYPE html>`
- `<html lang="en">`

```

• <head>
•   <meta charset="UTF-8">
•   <meta name="viewport" content="width=device-width, initial-scale=1.0">
•   <title>Premium Product Page | Online Store</title>
•   <link href="https://cdn.jsdelivr.net/npm/remixicon@3.5.0/fonts/remixicon.css"
rel="stylesheet">
•
•   <style>
•     /* --- BEM METHODOLOGY STYLING --- */
•     :root {
•       --color-primary: #2563eb;
•       --color-dark: #1e293b;
•       --color-light: #f8fafc;
•       --color-accent: #f59e0b;
•       --shadow: 0 10px 25px -5px rgba(0, 0, 0, 0.1);
•     }
•
•     * {
•       margin: 0;
•       padding: 0;
•       box-sizing: border-box;
•       font-family: 'Inter', sans-serif;
•     }
•
•     body {
•       background-color: #f1f5f9;
•       display: flex;
•       justify-content: center;
•       align-items: center;
•       min-height: 100vh;
•       padding: 20px;
•     }
•
•     /* BLOCK: Product Card */
•     .product-card {
•       background: white;
•       max-width: 900px;
•       width: 100%;
•       display: flex; /* Flexbox for responsive side-by-side layout */
•       border-radius: 20px;
•       overflow: hidden;
•       box-shadow: var(--shadow);
•       transition: transform 0.3s ease;
•     }
•
•     /* ELEMENT: Product Card Image Section */
•     .product-card__image-wrapper {
•       flex: 1;
•       background-color: #e2e8f0;
•       display: flex;

```

```

•         justify-content: center;
•         align-items: center;
•         padding: 40px;
•         position: relative;
•     }
•
•     .product-card__image {
•         max-width: 100%;
•         height: auto;
•         object-fit: contain;
•         transition: transform 0.5s ease;
•     }
•
•     /* MODIFIER: Hover effect on image */
•     .product-card:hover .product-card__image {
•         transform: scale(1.1) rotate(-2deg);
•     }
•
•     /* ELEMENT: Content Section */
•     .product-card__content {
•         flex: 1;
•         padding: 50px;
•         display: flex;
•         flex-direction: column;
•         justify-content: center;
•     }
•
•     .product-card__category {
•         text-transform: uppercase;
•         font-size: 0.8rem;
•         font-weight: 700;
•         color: var(--color-primary);
•         letter-spacing: 1px;
•         margin-bottom: 10px;
•     }
•
•     .product-card__title {
•         font-size: 2.5rem;
•         color: var(--color-dark);
•         line-height: 1.1;
•         margin-bottom: 15px;
•     }
•
•     .product-card__description {
•         color: #64748b;
•         margin-bottom: 25px;
•         font-size: 1rem;
•     }
•
•     .product-card__price-tag {

```

```

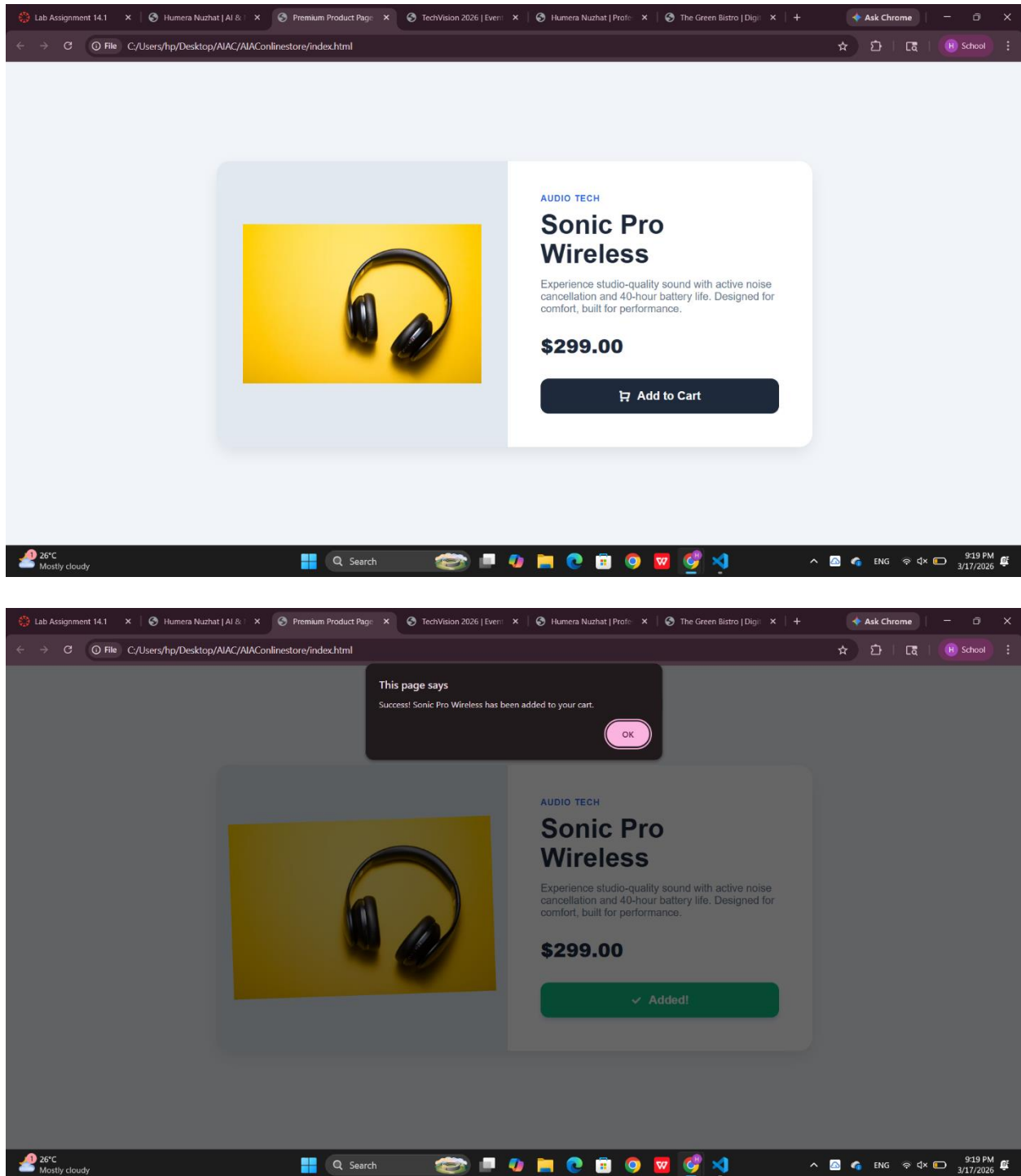
•         font-size: 1.8rem;
•         font-weight: 800;
•         color: var(--color-dark);
•         margin-bottom: 30px;
•     }
•
•     /* ELEMENT: Button */
•     .product-card__button {
•         background-color: var(--color-dark);
•         color: white;
•         border: none;
•         padding: 16px 32px;
•         border-radius: 12px;
•         font-size: 1.1rem;
•         font-weight: 600;
•         cursor: pointer;
•         display: flex;
•         align-items: center;
•         justify-content: center;
•         gap: 10px;
•         transition: all 0.3s ease;
•     }
•
•     /* MODIFIER: Button Hover */
•     .product-card__button:hover {
•         background-color: var(--color-primary);
•         box-shadow: 0 10px 15px -3px rgba(37, 99, 235, 0.4);
•         transform: translateY(-2px);
•     }
•
•     .product-card__button:active {
•         transform: translateY(0);
•     }
•
•     /* --- RESPONSIVE LAYOUT --- */
•     @media (max-width: 768px) {
•         .product-card {
•             flex-direction: column; /* Stack image on top of text on mobile */
•         }
•
•         .product-card__content {
•             padding: 30px;
•         }
•
•         .product-card__title {
•             font-size: 1.8rem;
•         }
•     }
•     </style>
• </head>

```

```

• <body>
•
•   <main class="product-card">
•
•     <div class="product-card__image-wrapper">
•       
•     </div>
•
•     <div class="product-card__content">
•       <span class="product-card__category">Audio Tech</span>
•       <h1 class="product-card__title">Sonic Pro Wireless</h1>
•       <p class="product-card__description">
•         Experience studio-quality sound with active noise cancellation and
40-hour battery life. Designed for comfort, built for performance.
•       </p>
•       <div class="product-card__price-tag">$299.00</div>
•
•       <button class="product-card__button" id="addToCartBtn">
•         <i class="ri-shopping-cart-line"></i> Add to Cart
•       </button>
•     </div>
•   </main>
•
•   <script>
•     // JS Requirement: Add to Cart Alert
•     const cartBtn = document.getElementById('addToCartBtn');
•
•     cartBtn.addEventListener('click', () => {
•       // Visual feedback on the button
•       cartBtn.innerHTML = '<i class="ri-check-line"></i> Added!';
•       cartBtn.style.backgroundColor = '#10b981'; // Success Green
•
•       // Professional Alert
•       setTimeout(() => {
•         alert('Success! Sonic Pro Wireless has been added to your cart.');
```

## OUTPUT:



### Task 3 – Event Registration Form

Build an event registration form for a conference.

## Requirements:

- Collect name, email, phone number, and session selection.
- Use AI to:
  - o Add form validation with JavaScript.
  - o Make the form accessible with labels and ARIA.
  - o Style with a professional look

**Prompt:** "Build a professional event registration form for a tech conference.

- **Fields:** Full Name, Email, Phone Number, and a dropdown for Session Selection.
- **Accessibility:** Ensure the form is accessible by using proper `<label>` tags and **ARIA** attributes (aria-required, aria-live, aria-describedby).
- **Validation:** Write a JavaScript function to validate that the email is in the correct format and the phone number is 10 digits before allowing submission.
- **Style:** Use a clean, corporate look with clear error messages in red."

```
• <!DOCTYPE html>
• <html lang="en">
• <head>
•   <meta charset="UTF-8">
•   <meta name="viewport" content="width=device-width, initial-scale=1.0">
•   <title>TechVision 2026 | Event Registration</title>
•   <style>
•     /* --- PROFESSIONAL THEME --- */
•     :root {
•       --primary-blue: #1e40af;
•       --error-red: #dc2626;
•       --success-green: #16a34a;
•       --input-border: #cbd5e1;
•       --bg-light: #f8fafc;
•     }
•
•     * {
•       box-sizing: border-box;
•       margin: 0;
•       padding: 0;
•       font-family: 'Segoe UI', system-ui, sans-serif;
•     }
•
•     body {
•       background-color: var(--bg-light);
•       display: flex;
•       justify-content: center;
•       align-items: center;
•       min-height: 100vh;
```



```

•         padding: 20px;
•     }
•
•     .form-container {
•         background: white;
•         padding: 40px;
•         border-radius: 12px;
•         box-shadow: 0 10px 25px rgba(0,0,0,0.05);
•         max-width: 500px;
•         width: 100%;
•     }
•
•     header {
•         text-align: center;
•         margin-bottom: 30px;
•     }
•
•     header h1 {
•         color: #0f172a;
•         font-size: 1.8rem;
•         margin-bottom: 8px;
•     }
•
•     header p {
•         color: #64748b;
•         font-size: 0.9rem;
•     }
•
•     /* --- ACCESSIBLE FORM ELEMENTS --- */
•     .form-group {
•         margin-bottom: 20px;
•         position: relative;
•     }
•
•     label {
•         display: block;
•         margin-bottom: 8px;
•         font-weight: 600;
•         color: #334155;
•         font-size: 0.9rem;
•     }
•
•     input, select {
•         width: 100%;
•         padding: 12px;
•         border: 1px solid var(--input-border);
•         border-radius: 6px;
•         font-size: 1rem;
•         transition: all 0.3s ease;
•         background-color: white;

```

```

    }
    input:focus, select:focus {
        outline: none;
        border-color: var(--primary-blue);
        box-shadow: 0 0 0 3px rgba(30, 64, 175, 0.1);
    }

    /* Error States */
    .form-group.error input {
        border-color: var(--error-red);
    }

    .error-message {
        color: var(--error-red);
        font-size: 0.8rem;
        margin-top: 5px;
        display: none; /* Hidden by default */
    }

    .form-group.error .error-message {
        display: block;
    }

    .submit-btn {
        width: 100%;
        background-color: var(--primary-blue);
        color: white;
        padding: 14px;
        border: none;
        border-radius: 6px;
        font-size: 1rem;
        font-weight: 600;
        cursor: pointer;
        transition: background 0.3s;
        margin-top: 10px;
    }

    .submit-btn:hover {
        background-color: #1e3a8a;
    }

    .submit-btn:disabled {
        background-color: #94a3b8;
        cursor: not-allowed;
    }
</style>
</head>
<body>

```

```

•   <div class="form-container">
•       <header>
•           <h1>Conference Registration</h1>
•           <p>Secure your spot for TechVision 2026</p>
•       </header>
•
•       <form id="registrationForm" novalidate aria-label="Event Registration
Form">
•
•           <div class="form-group" id="nameGroup">
•               <label for="fullName">Full Name</label>
•               <input type="text" id="fullName" name="fullName"
•                   placeholder="e.g. Humera Nuzhat"
•                   aria-required="true"
•                   aria-describedby="nameError">
•               <span class="error-message" id="nameError" aria-
live="polite">Please enter your full name.</span>
•           </div>
•
•           <div class="form-group" id="emailGroup">
•               <label for="email">Email Address</label>
•               <input type="email" id="email" name="email"
•                   placeholder="name@example.com"
•                   aria-required="true"
•                   aria-describedby="emailError">
•               <span class="error-message" id="emailError" aria-
live="polite">Please enter a valid email address.</span>
•           </div>
•
•           <div class="form-group" id="phoneGroup">
•               <label for="phone">Phone Number</label>
•               <input type="tel" id="phone" name="phone"
•                   placeholder="+91 0000000000"
•                   aria-required="true"
•                   aria-describedby="phoneError">
•               <span class="error-message" id="phoneError" aria-
live="polite">Please enter a 10-digit phone number.</span>
•           </div>
•
•           <div class="form-group" id="sessionGroup">
•               <label for="session">Select a Session</label>
•               <select id="session" name="session" aria-required="true" aria-
describedby="sessionError">
•                   <option value="">-- Choose a session --</option>
•                   <option value="ai-ml">Artificial Intelligence & ML</option>
•                   <option value="web-tech">Advanced Web Technologies</option>
•                   <option value="cloud">Cloud Computing & Dev Ops</option>
•               </select>
•               <span class="error-message" id="sessionError" aria-
live="polite">Please select a session to attend.</span>

```

```

    </div>
    <button type="submit" class="submit-btn">Register Now</button>
  </form>
</div>

<script>
  const form = document.getElementById('registrationForm');

  form.addEventListener('submit', (e) => {
    e.preventDefault();
    let isValid = true;

    // Reset previous errors
    document.querySelectorAll('.form-group').forEach(group =>
group.classList.remove('error'));

    // 1. Name Validation
    const nameInput = document.getElementById('fullName');
    if (nameInput.value.trim().length < 2) {
      showError('nameGroup');
      isValid = false;
    }

    // 2. Email Validation (Regex)
    const emailInput = document.getElementById('email');
    const emailPattern = /^[^\s@]+@[^\s@]+\.[^\s@]+$/;
    if (!emailPattern.test(emailInput.value)) {
      showError('emailGroup');
      isValid = false;
    }

    // 3. Phone Validation (Simple 10 digit)
    const phoneInput = document.getElementById('phone');
    if (phoneInput.value.trim().length < 10) {
      showError('phoneGroup');
      isValid = false;
    }

    // 4. Session Validation
    const sessionInput = document.getElementById('session');
    if (sessionInput.value === "") {
      showError('sessionGroup');
      isValid = false;
    }

    if (isValid) {
      alert('Success! Registration submitted for ' + nameInput.value);
      form.reset();
    }
  });

```

```
•     });  
•  
•     function showError(groupId) {  
•         document.getElementById(groupId).classList.add('error');  
•     }  
•     </script>  
• </body>  
• </html>
```

**Conference Registration**  
Secure your spot for TechVision 2026

Full Name

Email Address

Phone Number

Select a Session

[Register Now](#)

20°C Mostly cloudy  
Search  
9:19 PM 3/17/2026

## Conference Registration

Secure your spot for TechVision 2026

Full Name

Humera Nuzhat

Email Address

2303a52083@sru.edu.in

Phone Number

91000000

Select a Session

-- Choose a session --

-- Choose a session --

Artificial Intelligence & ML

Advanced Web Technologies

Cloud Computing & Dev Ops

## Conference Registration

Secure your spot for TechVision 2026

Full Name

Humera Nuzhat

Email Address

2303a52083@sru.edu.in

Phone Number

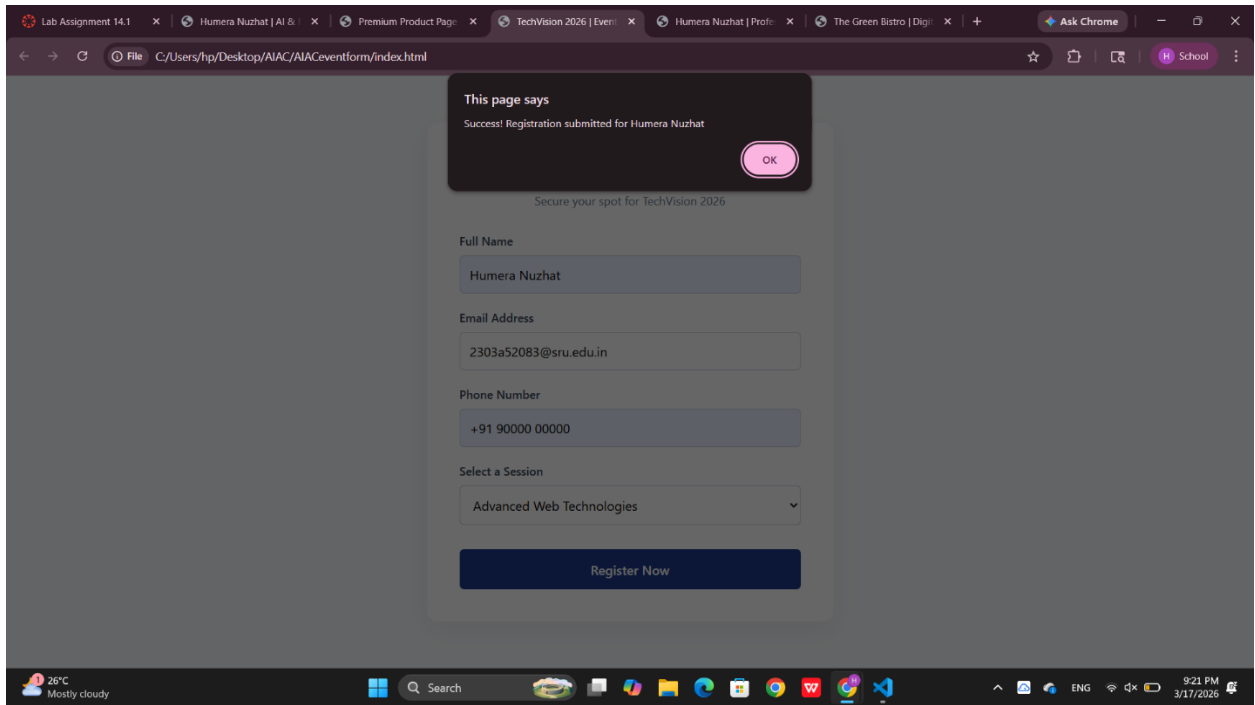
9100000000

Please enter a 10-digit phone number.

Select a Session

Artificial Intelligence & ML

Register Now



## Task 4 – Resume Website

Build a one-page online resume.

Requirements:

- Sections: Profile Summary, Skills, Education, Experience.
- Use AI to:
  - o Propose professional typography styles.
  - o Align sections using CSS Grid.
  - o Add smooth scroll navigation between sections.

**Prompt:** "Generate a one-page online resume using a sophisticated 'Executive' layout.

- **Typography:** Suggest a pairing of 'Playfair Display' for headings and 'Inter' for body text.
- **Layout:** Use **CSS Grid** to create a two-column layout (Sidebar for Skills/Education, Main for Summary/Experience).
- **Navigation:** Create a sticky header with links that use smooth-scroll to jump to specific sections.
- **Responsive:** Ensure the grid collapses to a single column on screens smaller than 768px."

```

• <!DOCTYPE html>
• <html lang="en">
• <head>
•   <meta charset="UTF-8">
•   <meta name="viewport" content="width=device-width, initial-scale=1.0">
•   <title>Humera Nuzhat | Professional Resume</title>
•
•   <link
href="https://fonts.googleapis.com/css2?family=Inter:wght@300;400;600&family=Playf
air+Display:wght@700&display=swap" rel="stylesheet">
•   <link href="https://cdn.jsdelivr.net/npm/remixicon@3.5.0/fonts/remixicon.css"
rel="stylesheet">
•
•   <style>
•     :root {
•       --primary: #1e293b; /* Slate 800 */
•       --accent: #0f172a; /* Slate 900 */
•       --text: #334155; /* Slate 600 */
•       --light-bg: #f8fafc;
•       --border: #e2e8f0;
•     }
•
•     * {
•       margin: 0;
•       padding: 0;
•       box-sizing: border-box;
•     }
•
•     html {
•       scroll-behavior: smooth;
•     }
•
•     body {
•       font-family: 'Inter', sans-serif;
•       color: var(--text);
•       background-color: #ffffff;
•       line-height: 1.7;
•     }
•
•     /* --- STICKY NAVIGATION --- */
•     .resume-nav {
•       position: sticky;
•       top: 0;
•       background: white;
•       border-bottom: 1px solid var(--border);
•       z-index: 100;
•       padding: 1rem 0;
•     }
•
•     .nav-container {

```



```

•         max-width: 1000px;
•         margin: 0 auto;
•         display: flex;
•         justify-content: center;
•         gap: 2rem;
•     }
•
•     .nav-container a {
•         text-decoration: none;
•         color: var(--primary);
•         font-weight: 600;
•         font-size: 0.9rem;
•         text-transform: uppercase;
•         letter-spacing: 1px;
•         transition: 0.3s;
•     }
•
•     .nav-container a:hover {
•         color: #6366f1;
•     }
•
•     /* --- CSS GRID MAIN LAYOUT --- */
•     .resume-wrapper {
•         max-width: 1000px;
•         margin: 40px auto;
•         display: grid;
•         grid-template-columns: 1fr 2fr; /* Sidebar 1 part, Content 2 parts */
•         gap: 60px;
•         padding: 0 20px;
•     }
•
•     h1, h2, h3 {
•         font-family: 'Playfair Display', serif;
•         color: var(--accent);
•     }
•
•     .profile-header {
•         grid-column: 1 / -1; /* Header spans full width */
•         text-align: center;
•         padding-bottom: 40px;
•         border-bottom: 2px solid var(--primary);
•         margin-bottom: 40px;
•     }
•
•     .profile-header h1 { font-size: 3.5rem; margin-bottom: 10px; }
•
•     /* --- SECTIONS --- */
•     .section-title {
•         font-size: 1.5rem;
•         margin-bottom: 20px;

```

```

•         display: flex;
•         align-items: center;
•         gap: 10px;
•         border-bottom: 1px solid var(--border);
•         padding-bottom: 5px;
•     }
•
•     .sidebar-section {
•         margin-bottom: 40px;
•     }
•
•     .skills-list {
•         list-style: none;
•     }
•
•     .skills-list li {
•         background: var(--light-bg);
•         margin-bottom: 8px;
•         padding: 8px 15px;
•         border-radius: 4px;
•         font-size: 0.9rem;
•         border-left: 3px solid var(--primary);
•     }
•
•     .experience-item, .education-item {
•         margin-bottom: 30px;
•     }
•
•     .date {
•         font-size: 0.85rem;
•         font-weight: 600;
•         color: #6366f1;
•     }
•
•     .job-title {
•         font-size: 1.2rem;
•         font-weight: 700;
•         color: var(--accent);
•     }
•
•     /* --- RESPONSIVE --- */
•     @media (max-width: 768px) {
•         .resume-wrapper {
•             grid-template-columns: 1fr; /* Stack into 1 column */
•         }
•         .profile-header h1 { font-size: 2.5rem; }
•         .nav-container { gap: 1rem; flex-wrap: wrap; }
•     }
•     </style>
• </head>

```

```

• <body>
•
•     <nav class="resume-nav">
•         <div class="nav-container">
•             <a href="#summary">Profile</a>
•             <a href="#skills">Skills</a>
•             <a href="#education">Education</a>
•             <a href="#experience">Experience</a>
•         </div>
•     </nav>
•
•     <div class="resume-wrapper">
•         <header class="profile-header">
•             <h1>Humera Nuzhat</h1>
•             <p>Artificial Intelligence & Machine Learning Student</p>
•         </header>
•
•         <aside class="sidebar">
•             <section id="skills" class="sidebar-section">
•                 <h2 class="section-title"><i class="ri-tools-fill"></i>
Skills</h2>
•                 <ul class="skills-list">
•                     <li>Python & C Programming</li>
•                     <li>Machine Learning Algorithms</li>
•                     <li>Database Management (SQL)</li>
•                     <li>Frontend (HTML/CSS/JS)</li>
•                     <li>Data Visualization</li>
•                 </ul>
•             </section>
•
•             <section id="education" class="sidebar-section">
•                 <h2 class="section-title"><i class="ri-book-open-fill"></i>
Education</h2>
•                 <div class="education-item">
•                     <p class="date">2023 - Present</p>
•                     <p class="job-title">B.Tech in CSE (AI & ML)</p>
•                     <p>SR University</p>
•                 </div>
•             </section>
•         </aside>
•
•         <main class="content">
•             <section id="summary" class="sidebar-section">
•                 <h2 class="section-title"><i class="ri-user-4-fill"></i> Profile
Summary</h2>
•                 <p>
•                     Ambitious Computer Science student at SR University with a
specialized focus on AI/ML.
•                     Proven ability to develop responsive web applications and
backend systems. Passionate

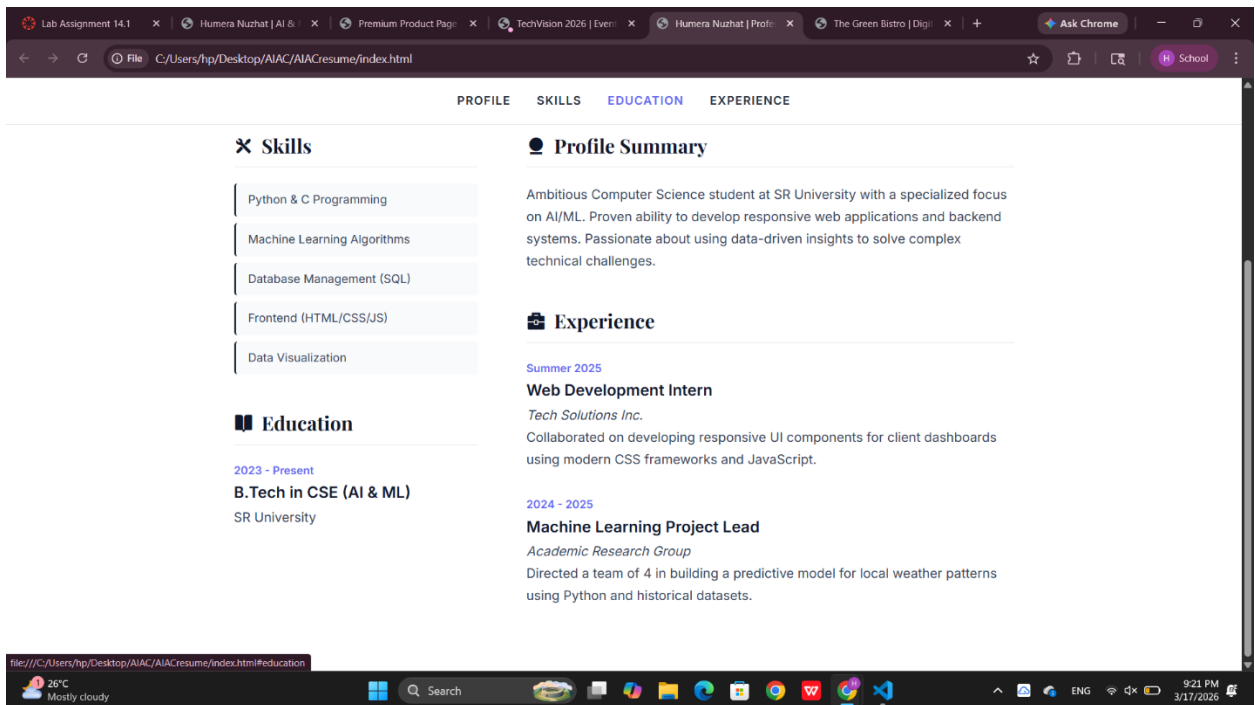
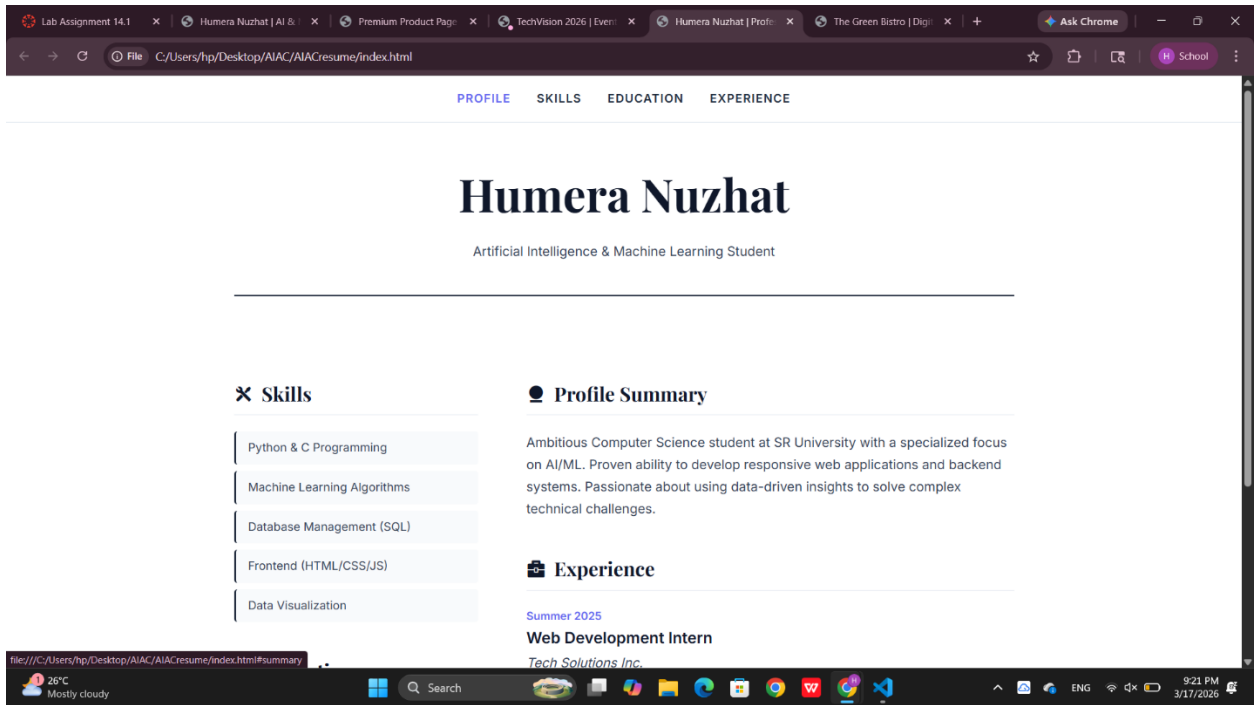
```

```

•         about using data-driven insights to solve complex technical
challenges.
•     </p>
•     </section>
•
•     <section id="experience" class="sidebar-section">
•         <h2 class="section-title"><i class="ri-briefcase-4-fill"></i>
Experience</h2>
•
•         <div class="experience-item">
•             <p class="date">Summer 2025</p>
•             <p class="job-title">Web Development Intern</p>
•             <p><em>Tech Solutions Inc.</em></p>
•             <p>Collaborated on developing responsive UI components for
client dashboards using modern CSS frameworks and JavaScript.</p>
•         </div>
•
•         <div class="experience-item">
•             <p class="date">2024 - 2025</p>
•             <p class="job-title">Machine Learning Project Lead</p>
•             <p><em>Academic Research Group</em></p>
•             <p>Directed a team of 4 in building a predictive model for
local weather patterns using Python and historical datasets.</p>
•         </div>
•     </main>
• </main>
• </div>
•
• <script>
•     // Smooth scrolling is handled via CSS "scroll-behavior: smooth"
•     // This JS ensures the URL updates without a jump
•     document.querySelectorAll('.nav-container a').forEach(link => {
•         link.addEventListener('click', (e) => {
•             console.log('Navigating to section...');
•         });
•     });
• </script>
• </body>
• </html>

```

**OUTPUT:**



## Task 5 – Restaurant Menu Page

Design a digital menu page for a restaurant.

Requirements:

- Display menu items with name, description, and price.

- Use AI to:

- o Suggest a food-friendly color palette.

- o Design a responsive card-based layout.

- o Add hover effects on menu items.

**Prompt:** "Design a digital menu for 'The Green Bistro' restaurant.

- **Palette:** Suggest a food-friendly color palette using forest greens, creams, and terracotta oranges.
- **Layout:** Create a responsive card-based layout using CSS Grid `auto-fit`.
- **Content:** Each card must show a food image, name, description, and price.
- **Hover:** Add a 'lift' animation and shadow increase when a user hovers over a menu card.
- **Font:** Use a high-end Serif font for the menu titles."

```
• <!DOCTYPE html>
• <html lang="en">
• <head>
•   <meta charset="UTF-8">
•   <meta name="viewport" content="width=device-width, initial-scale=1.0">
•   <title>The Green Bistro | Digital Menu</title>
•   <link href="https://cdn.jsdelivr.net/npm/remixicon@3.5.0/fonts/remixicon.css"
rel="stylesheet">
•   <link
href="https://fonts.googleapis.com/css2?family=Cormorant+Garamond:wght@600;700&family=Poppins:wght@300;400;500&display=swap" rel="stylesheet">
•
•   <style>
•     /* --- FOOD-FRIENDLY COLOR PALETTE --- */
•     :root {
•       --primary-green: #2d4a22; /* Forest green (Freshness) */
•       --warm-cream: #fdfaf5; /* Soft background */
•       --accent-orange: #d97706; /* Terracotta (Appetite) */
•       --text-dark: #1a1a1a;
•       --text-muted: #6b7280;
•     }
•
•     * {
•       margin: 0;
•       padding: 0;
•       box-sizing: border-box;
•     }
•
•     body {
•       font-family: 'Poppins', sans-serif;
•       background-color: var(--warm-cream);
```

```
•         color: var(--text-dark);
•         line-height: 1.6;
•     }
•
•     header {
•         text-align: center;
•         padding: 60px 20px;
•         background-color: var(--primary-green);
•         color: white;
•         clip-path: polygon(0 0, 100% 0, 100% 85%, 0% 100%);
•     }
•
•     header h1 {
•         font-family: 'Cormorant Garamond', serif;
•         font-size: 3.5rem;
•         margin-bottom: 10px;
•     }
•
•     .container {
•         max-width: 1100px;
•         margin: 40px auto;
•         padding: 0 20px;
•     }
•
•     /* --- RESPONSIVE CARD LAYOUT --- */
•     .menu-grid {
•         display: grid;
•         grid-template-columns: repeat(auto-fit, minmax(300px, 1fr));
•         gap: 30px;
•         margin-top: 40px;
•     }
•
•     .menu-item {
•         background: white;
•         border-radius: 15px;
•         overflow: hidden;
•         box-shadow: 0 4px 15px rgba(0,0,0,0.05);
•         transition: all 0.4s cubic-bezier(0.175, 0.885, 0.32, 1.275);
•         cursor: pointer;
•         border: 1px solid #f1f1f1;
•     }
•
•     /* --- HOVER EFFECTS --- */
•     .menu-item:hover {
•         transform: translateY(-10px);
•         box-shadow: 0 15px 30px rgba(45, 74, 34, 0.15);
•         border-color: var(--primary-green);
•     }
•
•     .menu-item_img {
```

```
•       width: 100%;  
•       height: 200px;  
•       object-fit: cover;  
•       transition: transform 0.5s ease;  
•   }  
•  
•   .menu-item:hover .menu-item__img {  
•       transform: scale(1.1);  
•   }  
•  
•   .menu-item__content {  
•       padding: 25px;  
•   }  
•  
•   .menu-item__header {  
•       display: flex;  
•       justify-content: space-between;  
•       align-items: center;  
•       margin-bottom: 10px;  
•   }  
•  
•   .menu-item__title {  
•       font-family: 'Cormorant Garamond', serif;  
•       font-size: 1.5rem;  
•       font-weight: 700;  
•       color: var(--primary-green);  
•   }  
•  
•   .menu-item__price {  
•       font-weight: 600;  
•       color: var(--accent-orange);  
•       font-size: 1.2rem;  
•   }  
•  
•   .menu-item__desc {  
•       font-size: 0.9rem;  
•       color: var(--text-muted);  
•       margin-bottom: 15px;  
•   }  
•  
•   .menu-item__tag {  
•       display: inline-block;  
•       background: #f3f4f6;  
•       padding: 4px 12px;  
•       border-radius: 20px;  
•       font-size: 0.75rem;  
•       font-weight: 500;  
•       color: var(--primary-green);  
•   }
```



```

•      /* Responsive adjustments */
•      @media (max-width: 600px) {
•          header h1 { font-size: 2.5rem; }
•          .menu-grid { grid-template-columns: 1fr; }
•      }
•      </style>
•  </head>
•  <body>
•
•      <header>
•          <h1>The Green Bistro</h1>
•          <p>Organic. Seasonal. Authentic.</p>
•      </header>
•
•      <main class="container">
•          <div class="menu-grid">
•
•              <article class="menu-item">
•                  
•                  <div class="menu-item__content">
•                      <div class="menu-item__header">
•                          <h3 class="menu-item__title">Avocado Zen Bowl</h3>
•                          <span class="menu-item__price">$14.50</span>
•                      </div>
•                      <p class="menu-item__desc">Fresh organic avocado, quinoa,
cherry tomatoes, and a light lemon-tahini dressing.</p>
•                      <span class="menu-item__tag"><i class="ri-leaf-line"></i>
Vegan</span>
•                  </div>
•              </article>
•
•              <article class="menu-item">
•                  
•                  <div class="menu-item__content">
•                      <div class="menu-item__header">
•                          <h3 class="menu-item__title">Wild Mushroom Pizza</h3>
•                          <span class="menu-item__price">$18.00</span>
•                      </div>
•                      <p class="menu-item__desc">Hand-stretched dough, truffle oil,
sautéed wild mushrooms, and fresh mozzarella.</p>
•                      <span class="menu-item__tag"><i class="ri-fire-line"></i>
Chef's Choice</span>
•                  </div>
•              </article>
•
•              <article class="menu-item">

```

```

•         
•         <div class="menu-item__content">
•             <div class="menu-item__header">
•                 <h3 class="menu-item__title">Aegean Greek Salad</h3>
•                 <span class="menu-item__price">$12.00</span>
•             </div>
•             <p class="menu-item__desc">Classic Mediterranean salad with
kalamata olives, feta cheese, and extra virgin olive oil.</p>
•             <span class="menu-item__tag"><i class="ri-seedling-line"></i>
Gluten Free</span>
•         </div>
•     </article>
•
• </div>
• </main>
•
• <script>
•     // Optional JS for a simple "Click to Order" feature
•     document.querySelectorAll('.menu-item').forEach(item => {
•         item.addEventListener('click', () => {
•             const title = item.querySelector('.menu-item__title').innerText;
•             alert(`Added ${title} to your order!`);
•         });
•     });
• </script>
•
• </body>
• </html>

```

**OUTPUT:**

